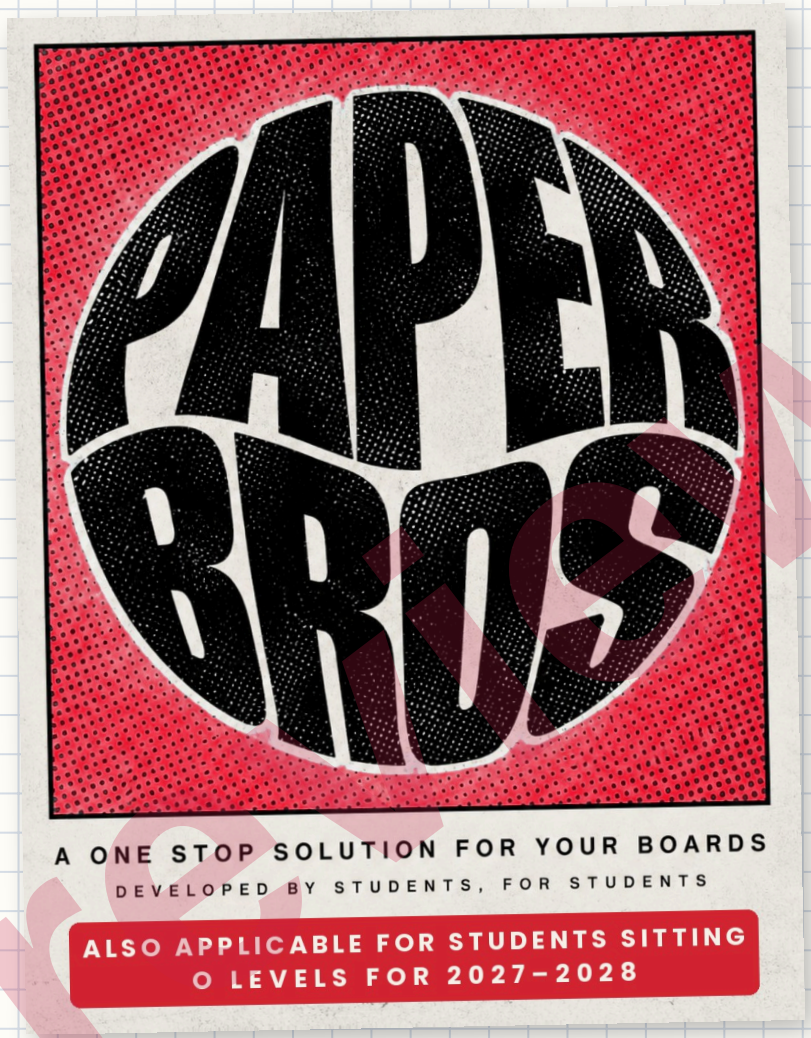
PREVIEW

Cambridge O Level

Economics 2281

the complete notes — for exams in 2027, 2028 & 2029



notes for every syllabus point · worked examples

accurate diagrams & graphs · Paper 1 & Paper 2 practice + answers

this notebook belongs to:

Name .....

School / class ..... Exam: June / Nov

Preview

PREVIEW

what's inside

Contents

tick the box when a topic is done & dusted ✓

#	Topic	What's inside (syllabus sections)	Notes pg	Practice pg	✓
–	**The exam at a glance + econ toolkit**	papers, marks, AOs, command words, how (d) answers are marked, maths you need, diagram rules	3	–	☐
1	**The basic economic problem**	1.1 scarcity, what/how/for whom, economic vs free goods · 1.2 factors of production · 1.3 opportunity cost · 1.4 PPC diagrams	5	10	☐
2	**The allocation of resources**	2.1 markets · 2.2 demand · 2.3 supply · 2.4 price determination · 2.5 price changes · 2.6 PED · 2.7 PES · 2.8 market system · 2.9 market failure · 2.10 mixed economy & intervention	12	36	☐
3	**Microeconomic decision-makers**	3.1 money & banking · 3.2 households · 3.3 workers · 3.4 firms · 3.5 firms & production · 3.6 costs, revenue & objectives · 3.7 types of markets	39	54	☐
4	**Government & the macroeconomy**	4.1 macro aims & conflicts · 4.2 fiscal policy · 4.3 monetary policy · 4.4 supply-side policy · 4.5 economic growth · 4.6 unemployment · 4.7 inflation	57	71	☐
5	**Economic development**	5.1 living standards (GDP per head, HDI) · 5.2 poverty · 5.3 population · 5.4 differences between countries	74	82	☐
6	**International trade & globalisation**	6.1 specialisation & free trade · 6.2 globalisation, MNCs & protection · 6.3 exchange rates · 6.4 current account	85	96	☐
*	**Answers & mark schemes**	Paper 1 keys + Paper 2 mark-scheme style answers (with level guidance for the 8-mark 'discuss' parts) for every topic	99	–	☐
*	**Last-minute pages**	must-know definitions, the diagram gallery, exam-day checklist & revision tracker	108	–	☐

Built on the 2027–2029 syllabus ✓

These notes follow Cambridge O Level Economics **2281** for exams in **2027, 2028 and 2029** — one syllabus covers all three years, so if you're sitting in 2027 or 2028 you're in exactly the right place. Every bullet in the syllabus has a home in here (look for the little grey [2.6.4] numbers).

How I'd use this book

1. read the notes → 2. cover them & redo the worked examples → 3. draw every diagram from memory → 4. do the practice without peeking → 5. mark with the answers at the back & fix your mistakes in red.

What's new in this syllabus (vs the old one)

Gone: trade unions as a separate topic (still in wages), growth of firms, principles of taxation, population pyramids, fixed exchange rates, the claimant count. **Added:** the concept of scarcity, perfectly inelastic / unitary / perfectly elastic, monopoly as a market failure, non-provision of public goods, arguments for & against the mixed economy, **age & culture** as influences on households, and **environmental sustainability** everywhere. All covered here.

Preview

PREVIEW

before we start...

The Exam at a Glance

You sit **two** papers. Both are compulsory and both test **all** of the syllabus. Grades go from **A*** down to **E**. Calculators are allowed in both.

Paper	Time	Marks	Weight	What it looks like	Tests
1 Multiple Choice	1 h	40	30%	40 compulsory four-option (A–D) questions — some need calculations or reading diagrams	AO1 + AO2
2 Structured Questions	2 h	80	70%	**Section A:** Q1 (20 marks) compulsory, based on a real-world data extract; parts (a)–(f) in the specimen are 2, 2, 2, 4, 4, 6 marks. **Section B:** answer **3 from 4** questions (Q2–Q5), 20 marks each, parts (a), (b), (c), (d)	AO1 + AO2 + AO3

AO Assessment objectives

Objective	All	P1	P2
AO1 Knowledge & understanding — definitions, formulas, concepts, theories, using econ words	43%	50%	40%
AO2 Analysis — select & interpret data, apply analysis to words, numbers, diagrams & graphs, build links	47%	50%	45%
AO3 Evaluation — judge arguments & data; see that decisions have *uncertain outcomes*	10%	0%	15%

So nearly **half** of everything is **analysis** — explaining *chains* of cause and effect. Learning definitions alone won't get you past a C.

✱ Paper 2 timing

120 min for 80 marks ≈ 1.5 min per mark. Section A ≈ 30 min, each Section B question ≈ 28 min, 6 min to check. In Section B the parts grow: **(a)** short define/identify ⟶ **(b)** explain ⟶ **(c)** analyse (often a diagram) ⟶ **(d)** discuss (8 marks, both sides + judgement).

key How to read these notes

- **maroon bold** = a key word that earns the mark
- yellow highlight = learn word-for-word
- pink / blue / green = colour-coded comparisons
- yellow box = **definition**; blue box = **worked example**
- sticky notes = exam tips & traps ⚠
- little grey numbers like [2.6.4] = the syllabus point

Paper 1 tactics

① ~1.5 min per question ② cover the options and answer it in your head first ③ cross out the 2 obviously wrong ones ④ for diagram questions, **sketch it** on the paper ⑤ watch for **NOT** / **always** / **most likely** ⑥ never leave a blank — no marks are lost for wrong answers.

cmd Command words

The first word tells you what the examiner wants. These are the ones the 2281 syllabus lists (plus one from the specimen):

Word	What you do
Define	give the **precise meaning** (learn the definitions in the yellow boxes)
State / Give	express clearly / produce an answer from the source or memory — short, no explanation
Identify	name / select / recognise (often from the extract)
Calculate	work it out from the figures — show working + units
Describe	state the points / give the main features (e.g. of a trend: up, down, peak, numbers)
Explain	give reasons: say *why* and/or *how*, make the links clear, back it up
Analyse	examine in detail: show the **chain** of cause ⟶ effect, often with a diagram
Discuss	write in depth in a structured way: **both sides**, then a **justified conclusion**
+ Draw	not in the syllabus list, but used in the specimen (e.g. 'Draw a diagram' in Section A): a fully labelled diagram

How a (d) 'discuss' answer gets 8/8

Section B (d) is marked in levels: **L3 6–8** both sides developed + reasoned discussion · **L2 3–5** clear analysis, thin or one-sided · **L1 1–2** bare statements. Section A (f) (6 marks) gives points: up to 4 each side.

① **Why yes** — 2 developed points (point ⟶ because ⟶ so...)

② **Why no** — 2 developed points the other way

③ **Conclusion** — 'it depends on...' + a reasoned judgement (size of change, time period, PED, how the money is spent, etc.). Repeating your points is not evaluation.

Top 3 mark-losers

① unlabelled diagrams (axes, curves, P1/Q1) ② 'it will increase' with no **because** ③ one-sided (d) answers or a conclusion that just repeats everything.

Preview

PREVIEW

the tools you'll use everywhere

Maths & Diagram Toolkit

5 calculations + 6 diagram rules = free marks in both papers

% The maths you need

Percentage change

$$\% \text{ change} = \frac{\text{new} - \text{old}}{\text{old}} \times 100$$

e.g. price rises from $8 to $10: (10 − 8) ÷ 8 × 100 = **+25%**. A fall gives a **minus** sign — keep it (it matters for PED!).

Percentage of a total (share)

$$\text{share} = \frac{\text{part}}{\text{total}} \times 100$$

Totals & averages

total cost = average cost × output | average = total ÷ number. Same trick for revenue (TR = P × Q), GDP per head (GDP ÷ population).

Index numbers

Base year = **100**. If CPI goes 100 ⟶ 104, prices rose **4%**. From 110 ⟶ 121 = (121 − 110)/110 = **10%** (not 11!).

Reading tables & charts

Check the **units** (millions? %? $bn?) and whether figures are **levels** or **rates of change**. A falling inflation rate (5% ⟶ 3%) still means prices are rising, just more slowly. Same for a falling growth rate.

Worked example ✎ *the classic trap*

GDP growth: 2025 = 4%, 2026 = 2%. Q: Did GDP fall in 2026?

A: **No** — GDP still **rose**, by 2%. It grew more slowly. GDP only falls when growth is **negative**.

Units & rounding

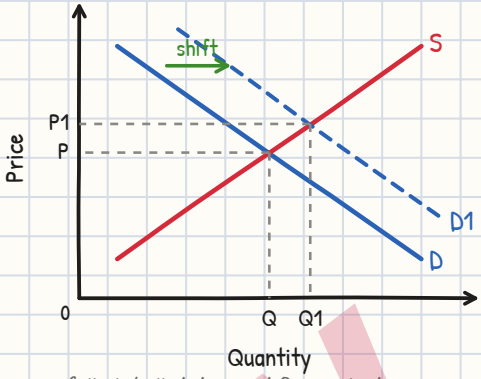
Always write the unit: **$**, **%**, **units**, **$m**. Round sensibly (usually 1 or 2 decimal places) and only at the **end**. A PED or PES has **no units** — it's just a number like −0.5 or 2.

percentage vs percentage point

Unemployment going from 4% to 6% is a rise of **2 percentage points**, but a **50% increase** in the rate ((6 − 4)/4 × 100).

Examiners love this one.

fig Diagram rules



a fully labelled demand & supply diagram

1. **Label both axes**: Price (vertical) and Quantity (horizontal). Labour market: Wage rate / Quantity of labour. FX market: price of currency in $ / quantity of currency.
2. **Label every curve**: D, S, then D1/S1 for the new one.
3. **Dashed lines** from each equilibrium to both axes, labelled P & Q, then P1 & Q1.
4. **Arrows** show the direction of a shift.
5. Draw it **big** (at least a third of a page) with a ruler — straight lines are fine.
6. Then **explain** the diagram in words: what shifts, why, new P & Q. A diagram alone rarely gets full marks.

Evaluation phrase bank ✎

'This depends on...' · 'In the short run... but in the long run...' · 'If demand is price inelastic, then...' · 'It may not work if...' · 'The most important factor is... because...' · 'Overall, on balance...'

Use the extract! In Section A, quote figures from the data ('rose from 3.1% to 4.6%') — answers that ignore the information can't get top marks.

Building a chain of analysis (the AO2 marks)

Every 'explain' or 'analyse' point should be a **chain**: cause ⟶ what happens in between ⟶ final effect. Each link usually earns a mark.

✗ weak: 'A tax on cigarettes reduces smoking.'

✓ strong: 'A tax on cigarettes **raises firms' costs**, so supply falls and the **price rises**; consumers buy fewer, so **consumption falls** — though if demand is **price inelastic** (addictive) the fall will be small.' ← that last bit is evaluation!

Preview

Topic 1

The Basic Economic Problem

too many wants, not enough stuff — so we have to choose

1.1 Finite resources, infinite wants

Economics is the study of **how we use scarce resources** to satisfy people's wants. Everything starts with one problem:

> **Basic economic problem:** resources are finite (limited) but wants are infinite (unlimited), so not every want can be satisfied and choices have to be made. [1.1.1]

> **Scarcity:** when there are not enough resources to produce everything that people want. Scarcity forces choice, and every choice has an opportunity cost.

needs vs wants? **Needs** = things required to survive (food, water, shelter, clothing). **Wants** = things people would like to have (a new phone, a holiday). Wants never end — get one and you want the next.

✱ The problem is everywhere [1.1.1]

Who?	Their limited resource	Example of the choice
Consumers	income / pocket money	new trainers **or** concert tickets with $80
Workers	time, skills, energy	which job? extra overtime **or** leisure? train now **or** earn now?
Producers / firms	land, workers, machines, finance	make more phones **or** tablets? buy a robot **or** hire staff?
Governments	tax revenue (and borrowing)	new hospital **or** new school **or** lower taxes?

exam tip ✎

Scarcity is about resources being limited **relative to wants** — even rich countries face it. Don't write 'scarcity means there is none left'.

scarcity → choice → opportunity cost

the chain behind <u>every</u> topic in this book — learn it in this order

Worked example ✎ *spot the problem*

Q: A government has $10bn to spend. Hospitals want $6bn, schools $5bn and new roads $3bn. Use this to explain the basic economic problem. [3]

A: Wants add up to $14bn but only $10bn is available [1] — resources are finite while wants are not, so there is **scarcity** [1]. The government must **choose** which to fund and which to cut back, and whatever it gives up is the opportunity cost [1].

✱ The 3 basic questions [1.1.2]

Because resources are scarce, **every** economy has to decide how to allocate them:

Question	What it means
What to produce?	which goods & services, and how much of each (more food or more cars? hospitals or weapons?)
How to produce?	which method / combination of resources — lots of workers (**labour-intensive**) or lots of machines (**capital-intensive**)?
For whom to produce?	who gets the output — those who can pay most? everyone equally? those in need?

Market economies answer these through the **price mechanism**; governments answer them in planned/mixed economies (Topic 2.4, 2.8, 2.10).

✱ Economic goods vs free goods [1.1.3]

	Economic good	Free good
scarce?	yes — uses scarce resources	no — unlimited supply
opportunity cost?	**yes** — making it means giving up something else	**zero** opportunity cost
price	usually has a price	no price (cannot charge)
examples	cars, bottled water, phones, a haircut, school lessons	air, sunlight, sea water, sand in a desert

Trap! Something given out <u>free of charge</u> is NOT a free good. State healthcare is 'free' to patients but uses doctors, medicines and tax money ⟶ it has an opportunity cost ⟶ it's an **economic good**.

Worked example ✎ *economic or free?*

Q: Is clean river water an economic good or a free good? Explain. [2]

A: In a remote area with plenty of rain it may be a **free good** — there's more than enough for everyone, so using it has no opportunity cost. In a city it is an **economic good**: it must be cleaned and piped using scarce resources (workers, chemicals, pipes) that could have been used for something else.

Preview

1.2 Factors of production

Factors of production: the **resources used to produce goods and services**: **land, labour, capital and enterprise. Each earns a reward (a factor income).** [1.2.1]

Factor	Definition (learn it!)	Examples	Reward
Land	**natural resources** – everything provided by nature that is used in production, on or under the ground or in the sea	fields, forests, rivers, fish, oil, gas, minerals, the site of a factory	rent
Labour	**human effort**, both physical and mental, used in producing goods and services	a teacher, a farm worker, a software engineer, a nurse	wages
Capital	**human-made goods used to produce other goods and services** (capital goods)	machines, tools, factories, delivery vans, computers, roads, ports	interest
Enterprise	the **willingness and ability to take risks** and to **organise** the other three factors to produce something	a person starting a bakery, the founder of a tech start-up	profit

capital ≠ money!

In economics, **capital** means **equipment** (a pizza oven), not cash in the bank. Money is only used to *buy* capital. Also: profit is a **reward for risk** – if the business fails, the entrepreneur makes a **loss**.

✱ What does an entrepreneur do?

- has the business idea and decides what, how and how much to produce
- hires labour, rents land, buys capital – **combines** the factors
- **bears the risk** of loss (uninsurable risk)
- innovates: new products, new methods

✱ Quick check: which factor?

tractor → **capital** · the farm's soil → **land** · the tractor driver → **labour** · the farmer who borrowed to start the farm and keeps the profit → **enterprise** · fish in the sea → **land** · the fishing boat → **capital** · a website designer's laptop → **capital**

Worked example ✎ factor rewards in a bakery

payment	$ per month	factor
shop rent	1 200	land
bakers' pay	3 000	labour
interest on oven loan	300	capital
what's left for the owner	?	enterprise

Sales revenue = $5 600. Q: Calculate the owner's reward.
A: 5 600 – (1 200 + 3 000 + 300) = $1 100 profit. If sales fell to $4 000 the owner would make a **loss** of $500 – that's the risk enterprise takes.

✱ Changes in quantity & quality [1.2.2]

The **quantity** = how much of a factor there is. The **quality** = how productive it is. More or better factors → an economy can produce more (the PPC shifts out – see 1.4).

	Quantity changes because...	Quality changes because...
Land	land reclamation; discovery of new oil/minerals ↑ · depletion of non-renewables, deforestation, rising sea levels, desertification ↓	fertilisers, irrigation, drainage ↑ · over-farming, soil erosion, pollution ↓
Labour	population growth, **immigration**, higher retirement age, more women joining the workforce (e.g. childcare) ↑ · emigration, ageing population, longer schooling ↓	**education & training**, better **healthcare** & diet, work experience ↑ · poor schooling, illness ↓
Capital	**investment** in new machines, factories & infrastructure ↑ · **depreciation** (wear and tear) not replaced ↓	**technological advance** – new machines are faster, more reliable ↑
Enterprise	lower taxes on profit, less red tape, easier loans, grants for start-ups, a culture that accepts risk ↑	business education & training, experience, mentoring ↑

sustainability link 🌱

Using up non-renewable land resources (oil, fish stocks, forests) faster than they are replaced reduces the quantity of land for **future** generations. That's why governments use quotas (Topic 2.10).

Mobile or not? Labour can move between jobs and places (Topic 3.3.4); land can't move at all, though it can change use (a farm becomes a housing estate). More mobile factors → resources shift faster to where they're wanted.

Preview

1.3 Opportunity cost

Opportunity cost: the next best alternative forgone (given up) when a choice is made. [1.3.1]

Only the **next best** option counts — not every option you didn't pick, and not the money spent.

Worked example ✎ ranking choices

Zara has $40. Her wish-list, in order: ① headphones ② a book ③ a meal out. She buys the headphones.

Q: What is the opportunity cost?

A: the book — her next best alternative. **Not** 'the book and the meal', and **not** '$40'.

✱ Opportunity cost in different contexts

Who	Choice	Opportunity cost
Student	stays at school until 18	the wages from a job they could have had
Farmer	plants rice on a field	the maize that field could have grown
Firm	spends profit on a new machine	e.g. paying higher dividends to owners or an advert campaign
Government	builds a motorway	e.g. the hospital that could have been built with the same money
Worker	works 10 hours overtime	10 hours of leisure (time with family)

free time isn't free

Even if you pay nothing, choices have an opportunity cost: going to a free concert costs you the **time** you could have spent revising. 😊

✱ Money cost vs opportunity cost

	Money (price) cost	Opportunity cost
measured in	$	the **real** thing given up (goods, time, leisure)
example	a cinema ticket costs $10	the pizza you could have bought with that $10 (if it was next best)
can it be zero?	yes (free entry)	only for **free goods** — time still has a cost

✱ Opportunity cost now vs the future

Cutting down a forest for timber today has an opportunity cost **for the future**: the forest's wood, wildlife and tourism are lost to later generations. Same for over-fishing or burning fossil fuels. Thinking about this trade-off is what environmental sustainability is all about.

✱ How opportunity cost shapes decisions [1.3.2]

Rational decision-makers compare the **benefit** of a choice with its **opportunity cost**. If the benefit is bigger, go for it.

Consumers — compare what they'd get from each product with limited income. A family that spends more on rent has less for holidays. A rise in the price of one product raises its opportunity cost (more of other things given up).

Workers — choosing a job means giving up the next best job (its wage + conditions). Going to university: OC = earnings forgone now, but likely higher wages later. Working longer hours: OC = leisure.

Producers / firms — using a factory and workers to make product A means not making product B. Firms switch resources to the product that gives the higher profit. Keeping profit to invest means less paid out to shareholders.

Governments — tax revenue spent on defence can't be spent on education. Cutting taxes may mean less to spend on public services. Governments must prioritise — e.g. a poor country may choose clean water before a new airport.

Worked example ✎ a 4-mark explain

Q: Explain the opportunity cost to a government of increasing spending on healthcare. [4]

A: The government has a limited budget (scarcity) [1]. Spending more on healthcare means less can be spent on other things [1], e.g. the next best alternative might be building new schools [1]. So the opportunity cost is the education services/schools forgone, which might reduce the quality of future labour [1] — or the government would have to raise taxes/borrow, and the OC falls on taxpayers' spending.

Worked example ✎ a firm's choice

A workshop can use its machines and workers this week to make **either** 120 chairs **or** 40 tables. Q: What is the opportunity cost of making one table?

A: 40 tables cost 120 chairs, so 1 table = 120 ÷ 40 = 3 chairs (and 1 chair = 1/3 of a table). If tables earn more than 3 chairs' worth of profit, the firm should make tables.

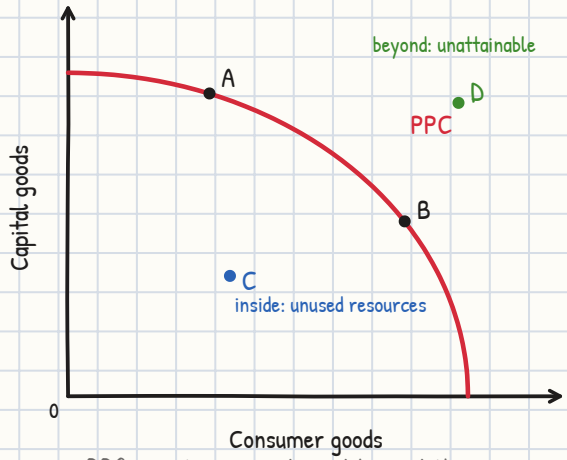
exam tip ✎

When asked for an opportunity cost, **name one specific** alternative ('the new school'), don't write 'other things' or 'money'. And link it back: 'so the opportunity cost is...'

Preview

1.4 Production possibility curves (PPC)

PPC: a curve showing the **maximum combinations of two goods (or services) that an economy can produce** when all its resources are fully and efficiently employed, with its current technology. [1.4.1]

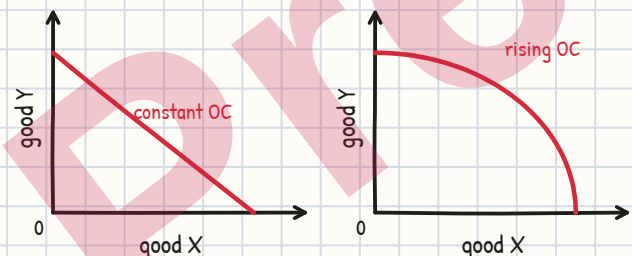


a PPC: points on, inside and beyond the curve

✱ Reading the diagram [1.4.2]

Point	Meaning
A, B (on)	all resources fully used — the economy is **productively efficient**; maximum output
C (inside)	**unemployed or underused resources** — inefficient, e.g. during a recession; more of **both** goods could be made with no opportunity cost
D (beyond)	**unattainable** with current resources & technology; can only be **produced** if the PPC shifts out (economic growth); trade can let a country **consume** beyond its PPC

✱ Straight-line vs curved PPC



straight = constant OC *curved = increasing OC*

a PPC assumes...

① only **two** goods are made ② resources and technology are **fixed** (in the short run) ③ resources on the curve are **fully & efficiently** used. Change ② and the whole curve shifts.

Worked example ✎ a straight-line PPC

An island can make 60 boats **or** 300 nets (or any straight-line mix).

Q: What is the opportunity cost of 1 boat? Of 10 boats?

A: 60 boats = 300 nets, so 1 boat = 5 nets; 10 boats = 50 nets.

Because the line is straight, the OC stays at 5 nets per boat wherever you are on it.

✱ Movements along a PPC [1.4.3]

Moving from A to B = **reallocating** resources from capital goods to consumer goods. To get more consumer goods, the economy **must give up** some capital goods ⟶ the PPC shows opportunity cost.

why curved? Resources are **not equally good** at making both goods. The first workers moved are the ones best suited to the new good; later ones are less suited, so **more and more** of the other good is given up for each extra unit ⟶ **increasing opportunity cost** ⟶ the curve bows outward (concave to the origin). A **straight-line** PPC means **constant** opportunity cost (resources equally suited).

Worked example ✎ OC from a PPC table

combo	V	W	X	Y	Z
robots (000s)	0	10	20	30	40
food (m tonnes)	100	95	80	55	20

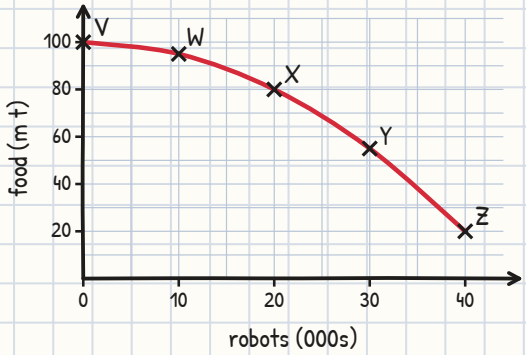
Q: Calculate the opportunity cost of producing 10 000 more robots when moving (i) W ⟶ X, (ii) Y ⟶ Z.

A: (i) 95 − 80 = 15 m tonnes of food (1 500 tonnes per robot)

(ii) 55 − 20 = 35 m tonnes of food (3 500 tonnes per robot)

The OC **rises** — that's why the PPC is curved.

(iii) Is 25 000 robots + 90 m tonnes of food possible? At 25 000 robots (between X and Y) the curve gives only about 68 m tonnes, so 90 is **beyond** the PPC ⟶ unattainable.



the table plotted — it gets steeper

draw it right ✎

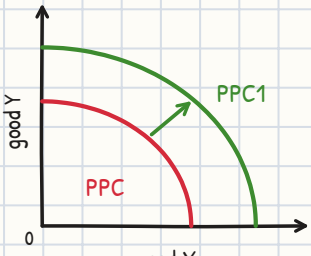
Plot from a table? Use a **ruler for the axes**, label them with **units** (robots in 000s, food in m tonnes), mark each point with a small **X**, then join with a **smooth curve**. Slope getting steeper = OC getting bigger.

Gradient = opportunity cost. On a PPC, the steeper the curve between two points, the more of the vertical-axis good you give up per extra unit of the horizontal-axis good.

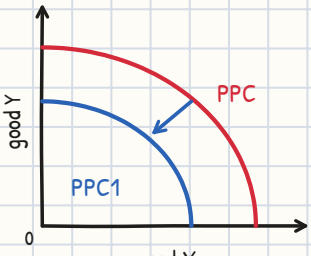
Preview

1.4 Shifts of a PPC

A **shift** happens when the **quantity or quality of resources** changes, or technology changes. [1.4.4]



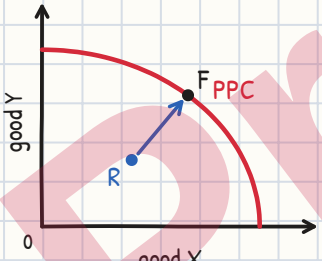
outward shift = growth



inward shift = lost capacity

Causes of an outward shift	Causes of an inward shift
• more labour: immigration, more women working, higher retirement age • better labour: education, training, healthcare • investment in more capital • new technology • discovery of natural resources, land reclamation	• natural disasters (floods, earthquakes) or war destroying capital • emigration of workers, ageing population • using up non-renewable resources • climate change, desertification • capital wearing out and not replaced

Economic growth (on a PPC): an outward shift = an increase in the economy's productive capacity (potential output). Moving from a point inside towards the curve uses idle resources — actual output rises but capacity doesn't.



recovery: R to F

In a **recession** output falls and workers lose jobs, so the economy sits at a point like R, **inside** the PPC. As it recovers, idle workers and machines are used again and it moves towards **F** on the curve. The PPC itself doesn't move.

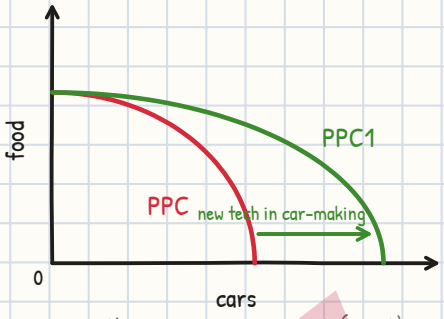
✱ Shift or movement? The cheat sheet

Event	On the PPC
unemployment rises in a recession	move **inside** the curve
idle factories re-open	move from inside **towards** the curve
workers move from farms to factories (full employment)	**movement along** the curve
new technology / more investment / immigration	**outward shift**
war, earthquake, emigration, resources used up	**inward shift**
better technology in one industry only	shift along **one axis** only

✱ Consequences of shifts

- **Outward**: more of **both** goods can be produced ⟶ higher potential output and living standards, more jobs possible, previously unattainable points (like D) become attainable.
- **Inward**: less can be produced ⟶ lower living standards, the economy has to cut output of something.

✱ Shifts for just one good



only the car axis moves out (pivot)

New technology **only in car production** ⟶ the PPC pivots outward along the car axis; maximum food output stays the same.

✱ Capital goods today = more growth tomorrow

An economy that puts more resources into **capital goods** now (factories, machines) gives up consumer goods today (**opportunity cost**) but its PPC will shift out **further in the future**. One that chooses mostly consumer goods enjoys more now, but grows more slowly.

exam tip ✎

A PPC diagram question wants: **labelled axes** (two goods), the **original** and **new** curve labelled (PPC, PPC1), and an **arrow**. Then explain the cause ⟶ what happened to resources ⟶ the effect on output. A fall in **unemployment** is not a shift — it's a move from inside the curve towards it!

Model answer ✎ 6-mark analyse

Q: Analyse, using a PPC diagram, how more spending on education could affect an economy. [6]

A: [diagram: axes = two goods [1]; PPC shifting outwards to PPC1, labelled, with arrow [1]] Education raises the **quality of labour** [1]: workers gain skills and become more productive [1]. With the same number of workers the economy can produce **more of both goods**, so productive capacity increases and the PPC shifts outwards [1]. Points that were unattainable become attainable, so potential output and living standards can rise [1].

(Caveats like 'it takes years' are evaluation — good for a (d), not credited in an 'analyse' (c).)

evaluate it: is an outward shift always good?

✓ more of both goods **can** be made. ✗ but **potential** output isn't **actual** output (a recession leaves you inside the new curve), and more output can mean more **pollution** — not sustainable.

Preview

Practice Questions

PREVIEW

PAPER 1 · MULTIPLE CHOICE

circle one answer – A, B, C or D

1 What is the basic economic problem?

A how to share income equally
B how to allocate finite resources to meet infinite wants
C how to produce goods at the lowest price
D how to reduce unemployment to zero

2 Which is a **free good**?

A tap water B sunlight C a state school D a public park

3 A firm buys a new delivery van. Which factor of production is the van?

A capital B enterprise C labour D land

4 What is the reward to enterprise?

A interest B profit C rent D wages

5 Ali can work an extra shift for $60, go to a match, or study. He chooses the match. His next best choice was to work. What is the opportunity cost of the match?

A the $60 he could have earned
B studying and the $60
C the price of the ticket
D nothing, he enjoyed it

6 Which would increase the **quality** of labour in a country?

A a rise in the birth rate
B more immigration of unskilled workers
C a government training scheme
D a higher retirement age

7 An economy is producing at a point inside its PPC. What does this show?

A the economy is fully employing its resources
B some resources are unemployed
C the output is currently unattainable
D there has been economic growth

8 Why is a PPC usually drawn curved (concave to the origin)?

A resources are not equally suited to producing both goods
B opportunity cost is constant
C there is unemployment
D wants are unlimited

9 Which would shift a country's PPC **inwards**?

A a fall in unemployment
B the discovery of new oil reserves
C a major earthquake destroying factories
D an increase in investment

10 A country can make 50 tractors and 200 tonnes of wheat, or 60 tractors and 170 tonnes of wheat. What is the opportunity cost of **one** extra tractor?

A 3 t B 10 t C 30 t D 170 t

11 Which question is part of the 'how to produce' decision?

A should we make more schools or more hospitals?
B should cars be made by robots or by workers?
C should the poor receive free medicine?
D how many phones should be made?

12 What is **always** true of an economic good?

A it is sold at a profit
B it is provided by the private sector
C it has an opportunity cost
D consumers must pay a price for it

Quick-fire recall ✦ *(say the answers out loud before you start Paper 2)*

1. Define scarcity & opportunity cost.
2. Four factors of production + their rewards?
3. What do points inside / on / beyond a PPC mean?
4. Why is a PPC usually curved?
5. Three causes of an outward shift, two of an inward one.
6. Free good vs economic good: one example each.

Preview

PAPER 2 · STRUCTURED

Section B style — 20 marks, about 28 minutes

1 Many countries are investing in renewable energy such as solar panels and wind farms. At the same time, some countries face falling numbers of workers as their populations age.

(a) Define *factors of production*. [2]

(b) Explain **two** reasons why the quantity of labour in a country may fall. [4]

(c) Analyse, using a production possibility curve diagram, the effect of an ageing population that reduces the number of workers in an economy. [6]

draw here

(d) Discuss whether a government should spend more of its budget on renewable energy rather than on healthcare. [8]

PAPER 2 · DATA RESPONSE

2 Read the extract and answer the questions.

Water scarcity in Country K

Country K is a lower-middle-income country where 60% of workers are farmers. Rainfall has fallen by 18% over ten years, and the river that feeds the main farming region is now often dry. The government spent $400m last year on new irrigation canals, and is considering a $250m desalination plant that turns sea water into drinking water. Some economists say the money would be better spent on schools, where only 55% of children finish secondary education.

(a) Identify the factor of production that irrigation canals are an example of, and state the reward to that factor. [2]

(b) Using the extract, explain why water is an economic good in Country K. [2]

(c) Explain the opportunity cost of building the desalination plant. [2]

(d) Explain how the fall in rainfall affects the quantity and quality of land in Country K. [4]

(e) Analyse how spending on schools could shift Country K's PPC. [4]

(f) Discuss whether Country K's government should build the desalination plant. [6]

mark yourself!

Paper 1 = 12 marks. Paper 2: Q1 = 20, Q2 = 20 ⟶ **52 marks**. Answers are at the back. Lost marks on the PPC? Redraw it three times — seriously, it sticks.

Preview

Topic 2

The Allocation of Resources

demand, supply, prices & elasticity — the engine room of economics

2.1 How markets work

Market: any place or arrangement where buyers and sellers come together to exchange goods and services (or factors of production). [2.1.1]

A market does **not** need a building. Buying a phone case on an app, a farmer's stall, a phone call to a supplier — all markets.

✱ Examples of markets [2.1.1]

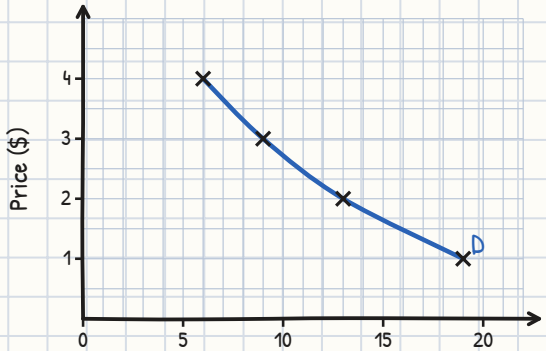
Market	What's traded
product markets	goods & services: rice, bubble tea, haircuts, phones, football tickets
labour market	workers' time & skills — the 'price' is the **wage** (Topic 3.3)
foreign exchange market	currencies, e.g. dollars for taka (Topic 6.3)
commodity markets	raw materials: oil, gold, wheat, cotton
online markets	shopping apps, ride-hailing, food delivery

✱ Roles of buyers and sellers [2.1.1]

	Buyers (consumers)	Sellers (producers / firms)
do what?	create **demand**	create **supply**
want...	the most satisfaction from their limited income (low prices, good quality)	usually the most **profit** (high prices, low costs)
signal by...	spending more or less — 'voting with their money'	making more or less, entering or leaving the market

Buyers and sellers want **opposite** things on price, and the market price is where their plans match up. That meeting point is **equilibrium** (2.4).

✱ Drawing the demand curve [2.2.1]



the market demand schedule plotted

2.2 Demand

Demand: the willingness and ability to buy a product at a given price, over a given period of time. [2.2.1]

wanting ≠ demand!

You might *want* a sports car, but with $15 in your pocket that's not demand. Economists only count demand **backed by money** (sometimes called **effective demand**).

✱ The law of demand

As the **price rises**, the **quantity demanded falls** (and vice versa), other things being equal. Why?

- people's income buys **less** when price rises, so they can afford fewer
- the product looks worse value than **substitutes**, so people switch
- some buyers are simply priced out of the market

✱ Individual & market demand [2.2.1]

Individual demand = one buyer's demand. **Market demand** = the total of all individual demands at each price. Just add across!

Worked example ✎ *adding up demand*

price ($)	Ana	Ben	Chen	market
4	2	3	1	**6**
3	3	4	2	**9**
2	4	6	3	**13**
1	6	8	5	**19**

Each column is a **demand schedule** (cups of bubble tea per week). At $2 the market demand = 4 + 6 + 3 = 13 cups. The market schedule is plotted on the left.

- Price on the **vertical** axis, quantity on the **horizontal**.
- Slopes **downwards** from left to right — that's the law of demand in a picture.
- Each point = how much buyers would buy at that price. Straight-line curves are fine in the exam.

Two little phrases that matter: demand is always **per period of time** (cups per week, cars per year) — '500 cups' means nothing without 'a day' or 'a week'. And **ceteris paribus** = 'other things being equal': when we link price and quantity demanded, we hold income, tastes and everything else constant.

Preview

2.2 Moving along the curve

A change in the product's **own price** causes a movement along the demand curve. The curve itself stays put. [2.2.2]

Term	Cause	Movement
extension in demand	price **falls** ⟶ quantity demanded rises	down the curve (A ⟶ B)
contraction in demand	price **rises** ⟶ quantity demanded falls	up the curve (B ⟶ A)



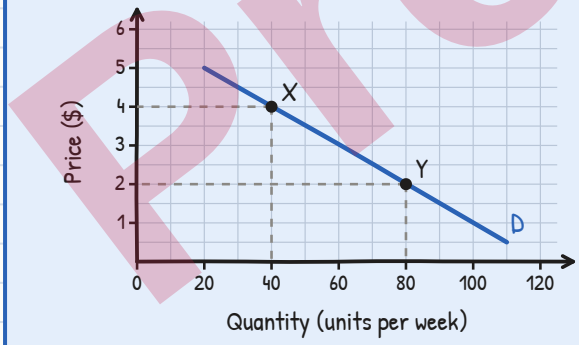
price falls P1 to P2: extension in demand

The **whole** demand curve D is still there — buyers have simply moved from point A to point B on it. Their tastes, incomes and everything else are unchanged.

why the movement happens

Cheaper bubble tea ⟶ your $10 buys more cups (income goes further) **and** it now looks good value next to juice or coffee (people switch towards it) ⟶ quantity demanded rises.

Worked example ✎ reading a demand diagram



Q: State the quantity demanded at $4 and describe what happens when the price falls to $2.

A: At $4, **40 units**. At $2, **80 units** — an extension in demand of 40 units, a movement from X to Y down the curve.

Bonus: spending = $4 × 40 = $160 and $2 × 80 = $160 — unchanged! (see PED & revenue, 2.6.4)

exam trap ⚠

'A rise in price causes demand to fall' is **wrong** wording. Price changes cause a fall in **quantity demanded** (a contraction). 'Demand falls' means the whole curve shifts left. Say it right, get the mark.

Worked example ✎ movement or shift?

Q: The price of cinema tickets rises from $8 to $10 and sales fall from 900 to 750 a week. Is this a shift of the demand curve?

A: **No.** The cinema's **own price** changed, so it's a movement up the demand curve — a contraction in demand of 150 tickets a week.

mnemonic **Own price = along. Anything else = shift.** Burn it into your brain; it decides half the diagram questions in Topic 2.

✱ Quick quiz: shift or movement? (apple market)

Event	Answer
the price of apples falls	**movement** — extension
the price of pears (a substitute) falls	**shift left** (decrease)
a TV doctor says 'an apple a day…'	**shift right** (increase)
incomes rise (apples are a normal good)	**shift right**
the price of apples rises after a frost	**movement** — contraction (the frost shifts *supply*)
caramel (a complement for toffee apples) gets cheaper	**shift right**

Cover the right-hand column and test yourself. 6/6 or go round again!

Things that DON'T shift demand: the product's **own price** (that's a movement), and anything that hits **producers** — costs, wages, weather on crops, taxes on firms, new technology. Those shift **supply**.

Joint & derived demand. Complements are in **joint demand** (bought together). **Derived demand** = demand for something because of the demand for *another* product: demand for bakers comes from demand for bread, demand for steel from demand for cars and buildings (more in 3.3 & 3.5).

Worked example ✎ Paper 1 style

What will cause an **extension** in demand for rice?

A a rise in incomes *B* a fall in the price of rice

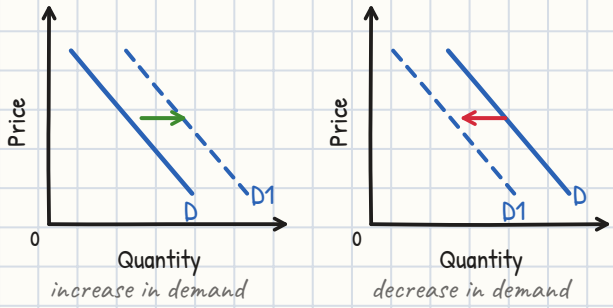
C a fall in the price of wheat *D* an advert for rice

Answer: B — only rice's own price causes a movement. A and D shift D right; C (a substitute gets cheaper) shifts it left.

Preview

2.2 Shifts of the demand curve

If something **other than the product's own price** changes, the **whole curve shifts**: more (or less) is demanded at **every** price. [2.2.3]



Increase in demand: more is demanded at each and every price — the curve shifts right. A decrease in demand shifts it left.

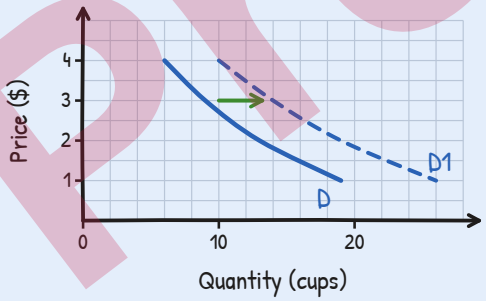
✱ Substitutes vs complements

	Substitutes	Complements
meaning	goods used **instead of** each other	goods used **together** (jointly demanded)
examples	tea & coffee, Pepsi & Coke, bus & train	cars & petrol, phones & apps, printers & ink
price of the other good rises →	demand for this good **increases** (people switch to it)	demand for this good **decreases** (fewer buy the pair)

Worked example ✎ a shift shown in a schedule

A heatwave hits. Market demand for bubble tea (cups per week):

price ($)	4	3	2	1
before (D)	6	9	13	19
heatwave (D1)	10	14	19	26



More is demanded at **every** price → the demand curve has shifted right, an increase in demand.

normal vs inferior goods

For most goods (**normal goods**) a rise in income → demand **increases**. For **inferior goods** (cheap own-brand noodles, second-hand clothes, bus travel for some) a rise in income → demand **decreases**, as people trade up.

✱ Causes of shifts in demand [2.2.3]

Change in...	Demand increases (right) if...
income	incomes rise (normal goods); incomes fall (inferior goods)
price of a substitute	the substitute gets **dearer**
price of a complement	the complement gets **cheaper**
tastes & fashion	it becomes trendy, a health report praises it, an influencer posts it
advertising	a successful campaign
population size & age structure	more people; more of the age group that buys it (ageing population → more care homes)
income tax	income tax is cut (more disposable income)
interest rates	rates fall — cheaper to borrow for cars, houses
seasons & weather	hot weather → ice cream, umbrellas in monsoon
expected future prices	people expect the price to rise soon (buy now!)

Reverse each one and demand **decreases** (shifts left).

Worked example ✎ explain a shift [4]

Q: Explain the effect of a fall in the price of petrol on the market for cars. [4]

A: Petrol and cars are **complements** [1]. Cheaper petrol makes owning a car cheaper to run [1], so **demand for cars increases** — the demand curve shifts to the right [1], causing a higher price and quantity of cars sold (if supply is unchanged) [1].

Worked example ✎ a decrease in demand

Q: Identify **two** reasons why demand for petrol cars might decrease.

A: ① electric cars (a **substitute**) become cheaper; ② **tastes** change as people worry about pollution and climate change. (Also: higher petrol prices — a complement; higher interest rates on car loans.) Either way, D shifts **left**.

the shift answer chain ✎

event → **why** buyers now want more/less → 'at **each** price' → D shifts right/left → new P and Q. Five links = full marks on most 4-mark 'explain' questions.

Population & age structure: a baby boom → more demand for nappies, then schools; an **ageing** population → more demand for healthcare and care homes; people moving to cities → more demand for urban housing and transport.

Preview

2.3 Supply

Supply: the **willingness and ability** of producers to sell a product at a given price, over a given period of time. [2.3.1]

✱ The law of supply

As the **price rises**, the **quantity supplied rises**, other things being equal, because:

- a higher price means more **profit** per unit — an incentive to make more
- firms can now cover the **higher costs** of producing extra units (overtime, extra machines)
- new firms are attracted into the market

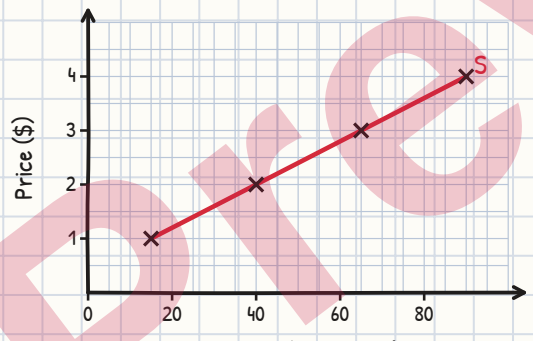
✱ Individual & market supply [2.3.1]

Market supply = the **total of all individual firms' supply** at each price.

Worked example ✎ *adding up supply*

price ($ per kg)	Farm X	Farm Y	Farm Z	market
1	10	5	0	**15**
2	20	12	8	**40**
3	28	20	17	**65**
4	35	27	28	**90**

Mango supply (kg per day). At $1, Farm Z supplies nothing — the price doesn't cover its costs. At $3: 28 + 20 + 17 = **65 kg**.



the market supply curve slopes upwards

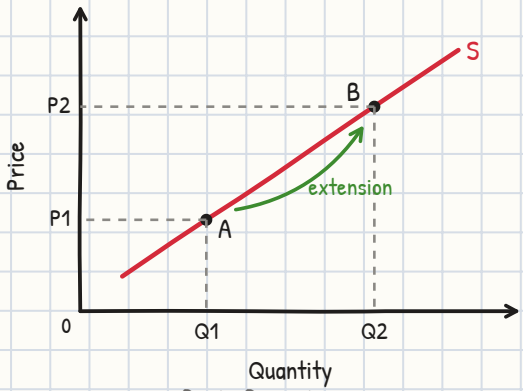
real life: durian fever

When durian prices boomed, Malaysian farmers swapped rubber and palm oil trees for durian trees. Higher price ⟶ more profit ⟶ more land, labour and capital moved into durians. That's the law of supply working over time.

Supply is a plan, not a pile. Supply is what firms are **willing and able to sell** at each price over a period — not the stock sitting in a warehouse.

✱ Movements along the supply curve [2.3.2]

A change in the product's **own price** ⟶ a **movement along** the supply curve.



price rises P1 to P2: extension in supply

Term	Cause	Movement
extension in supply	price **rises** ⟶ quantity supplied rises	up the curve (A ⟶ B)
contraction in supply	price **falls** ⟶ quantity supplied falls	down the curve (B ⟶ A)

spot the difference

Demand: price **down** = extension. Supply: price **up** = extension. Both 'extension' words mean **more** quantity — buyers want more when it's cheaper, sellers offer more when it's dearer. 🙂

Worked example ✎ *using the schedule*

Q: Using the mango table, describe what happens to supply when the price rises from $2 to $4.

A: Quantity supplied rises from 40 kg to 90 kg a day — an **extension in supply** of 50 kg, shown as a movement up the supply curve. The supply curve does **not** shift.

Supply schedule = a table showing the quantity supplied at each price. **Supply curve** = the same information drawn as a graph. Same idea for demand.

Paper 1 trap: 'A rise in the price of mangoes increases the supply of mangoes.' **Wrong** — it causes an **extension** in supply (a movement along). 'Supply increases' is saved for a **shift right**.

why not supply MORE?

Sometimes firms can't respond to a higher price straight away — no spare machines, crops still growing, no stock. How much quantity supplied responds is the **price elasticity of supply** (2.7).

Preview